

proposed a method called marginal Simulation-Based Calibration (marginal SBC) for Bayes factors (for an alternative see Sekulovski, Marsman, & Wagenmakers, 2024). Marginal SBC provides a means to assess whether Bayes factors are estimated accurately, or whether there is liberal or conservative bias in the Bayes factor estimator. In essence, marginal SBC works by defining a model prior (i.e., a priori model probabilities) and a parameter prior for each model (i.e., prior densities for each model parameter), and then doing a simulation many times, where in each simulation, one performs the following steps: (1) sample a model from the model prior, (2) for the sampled model, sample parameters from their prior distributions, (3) use the sampled parameters to simulate data from the model, (4) compute the Bayes factor using the method to be evaluated, and then use these Bayes factor estimates to compute the posterior model probabilities for each of the models given the simulated data. We can then compare the average posterior probabilities (averaged across simulations) with the prior probability; this allows marginal SBC to detect whether the Bayes factor estimates are biased or not. If the average posterior probability is equal to the prior probability, then this is consistent with accurate Bayes factor estimation. If the average posterior deviates from the prior, then this indicates a liberal or conservative bias in the Bayes factor estimation.

An important application domain of Bayes factor estimation is in the analysis of factorial designs using linear mixed-effects models (LMMs, Baayen, Davidson, & Bates, 2008). LMMs are now increasingly used in experimental psychology and related fields to study the outcome from experiments with factorial designs, and have developed into a standard approach, which often replaces previous ANOVA-based analyses. Nested model comparisons for balanced factorial designs can be analyzed using Bayes factors, which can replace traditional p-values as a means of assessing main effects, interactions, and specific contrasts (Schad, Vasishth, Hohenstein, & Kliegl, 2020). In this context, Bayes factors are often used for null hypothesis tests, where a full model containing all fixed effects of interest is compared to a null model that lacks some effect or a set of effects (van Doorn, Aust, Haaf,

Stefan, & Wagenmakers, 2021).

In this paper, we aim to investigate whether Bayes factor estimates obtained from common factorial designs are biased or accurate using marginal SBC as well as calibration plots that can be derived from it (Modrák & Bürkner, in preparation). We build upon our prior work (Schad, Nicenboim, & Vasishth, 2024), which showed that data aggregation can lead to biased inferences in Bayesian linear mixed models and ANOVA. To further assess the accuracy of Bayes factor estimation, we focus on a pair of widely used R packages: `bridgesampling` (Gronau et al., 2020) in combination with `brms` (Bürkner, 2017). Specifically, we apply marginal SBC to a range of experimental repeated measures designs, including a 2x2 design with random effects for subjects, a design with a single fixed factor and crossed random effects for subjects and items, and a 2x2 design with crossed random effects for subjects and items.

We focus on these designs here because they are the dominant type of experimental design in cognitive psychology and linguistics. The dominance of such designs and statistical tools in these areas is clear from the fact that tutorial papers like Pinheiro and Bates (2006), Baayen et al. (2008), and Bates, Mächler, Bolker, and Walker (2015) continue to be widely referenced in major journals like *Cognition* and the *Journal of Memory and Language* and in classic textbooks such as Gelman and Hill (2007) and Gelman et al. (2013). That linear mixed models continue to be a very important tool is clear from the steady stream of articles relating this approach that keep appearing in major psychology and psycholinguistic journals. A recent example illustrating the interest in mixed models is the special issue exclusively devoted to this topic in the official journal of the Society for Mathematical Psychology, *Computational Brain and Behavior*; see van Doorn et al. (2021).

Returning to our main goal in the present paper, by using marginal SBC, we can determine whether the Bayes factor estimates for the fixed effects are unbiased and provide a solid basis for statistical data analysis in these experimental designs. This investigation is

important, as biased Bayes factor estimates can lead to erroneous conclusions about the evidence for hypotheses and the relative strengths of models. Overall, our study contributes to the growing literature on the application of Bayesian methods in the cognitive sciences (psychology, linguistics, and related areas).

To foreshadow our results: we found that Bayes factor estimates for linear mixed models based on the R-packages `bridgesampling`/`brms` were accurate when no warning message was issued by the bridge sampler. However, when a warning message was issued, Bayes factor estimates could turn out biased and liberal, i.e., larger than the true Bayes factor in favor of H_1 . These results suggest that the accuracy of Bayes factor estimates in commonly used software tools cannot be taken for granted and that warning messages should be taken seriously. More generally, we suggest (cf. Schad et al., 2023) that marginal SBC for Bayes factors should be employed to test the accuracy for a given set of design, models, and priors before results from individual analyses are trusted.

Next, we provide a brief introduction to Bayesian analyses and Bayes factors, before we report the methods and results of our simulation studies.

Introduction to Bayesian analyses, Bayes factors, and simulation-based calibration

Here, we provide a brief introduction to Bayesian analyses, Bayes factors, and to marginal SBC for Bayes factors. This introduction is taken from Schad et al. (2024). Other introductory treatments are available for Bayesian and Bayes factor analyses (Chow & Hoijtink, 2017; Etz, Gronau, Dablander, Edelsbrunner, & Baribault, 2018; Etz & Vandekerckhove, 2018; Hoijtink & Chow, 2017; Lee, 2011; Mulder & Wagenmakers, 2016; Nicenboim, Schad, & Vasishth, 2025; Nicenboim & Vasishth, 2016; van Doorn et al., 2021; Vandekerckhove et al., 2018; Vasishth, Nicenboim, Beckman, Li, & Kong, 2018).